

Mathematics for Computer Science – QR

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Table of contents

- 1 Quadratic Residues.
- 2 Legendre Symbol and Euler's Criterion.
- 3 Quadratic Reciprocity.

Review: Finding the patterns.

Review.

Let $p = 7$, and for every $1 \leq a < p$, compute $a^i \bmod p$, where $1 \leq i < p$. We have:

a	a^2	a^3	a^4	a^5	a^6
1	1	1	1	1	1
2	4	1	2	4	1
3	2	6	4	5	1
4	2	1	4	2	1
5	4	6	2	3	1
6	1	6	1	6	1

May you find some patterns?

Review: Finding the patterns.

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a	a^2	a^3	a^4	a^5	a^6
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2	4	1	2	4	1
3	2	6	4	5	1
4	2	1	4	2	1
5	4	6	2	3	1
6	1	6	1	6	1

May you find some patterns?

Exercise.

总结倒数第二列的规律，陈述你的结论，并证明它。

First Question.

Given p be an odd prime and y an integer relatively prime to p , is y a perfect square modulo p ? Equivalently, we need to decide whether there exists $x \in \mathbb{Z}$ such that:

$$x^2 \equiv y \pmod{p}.$$

A Naïve Solution.

Easy jobs.

Let $p = 11$, and $y = 5$, is y a perfect square modulo p ? We write a simple program to square every number from 1 to 10. The output is:

1, 4, 9, 5, 3, 3, 5, 9, 4, 1

More data.

y	y^2
1	1
2	4
3	4
4	1

Table: $y^2 \bmod 5$

y	y^2
1	1
2	4
3	2
4	2
5	4
6	1

Table: $y^2 \bmod 7$

More data.

y	y^2
1	1
2	4
3	9
4	3
5	12
6	10
7	10
8	12
9	3
10	9
11	4
12	1

Table: $y^2 \bmod 13$

To find some patterns.

Pattern!

Can you find some patterns from the tables?

Some easy conclusions.

One easy conclusion.

$$(p-1)^2 \equiv 1 \pmod{p}.$$

Some easy conclusions.

One easy conclusion.

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Do you see why?

Some easy conclusions.

One easy conclusion.

$$(p-1)^2 \equiv 1 \pmod{p}.$$

Do you see why?

Easy proof.

It is easy to prove that:

$$(p-1)^2 = p^2 - 2p + 1 \equiv 1 \pmod{p}.$$

The pattern means that the congruence $x^2 \equiv 1 \pmod{p}$ has two solutions, 1 and $p-1$.

Some easy conclusions.

Another easy conclusion.

The second column of each table is symmetric under reflection (反射对称), as described by the following formula.

$$(p - a)^2 \equiv a^2 \equiv b \pmod{p}.$$

Some easy conclusions.

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The proof is similar, since:

$$(p - a)^2 = p^2 - 2pa + a^2.$$

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The proof is similar, since:

$$(p - a)^2 = p^2 - 2pa + a^2.$$

This formula means the congruence $x^2 \equiv b \pmod{p}$ has two incongruent solutions, a and $p - a$.

One formally stated property.

Proposition

Let p be an odd prime and b an integer not divisible by p . Then, the congruence

$$x^2 \equiv b \pmod{p}$$

has either no solutions or exactly two incongruent solutions modulo p .

One formally stated property.

Proof.

Assume that $x^2 \equiv b \pmod{p}$ has a solution. Then we must show that there are no more than two incongruent solutions. Assume that x_0 and x_1 are two solutions of $x^2 \equiv b \pmod{p}$. Then

$$x_0^2 \equiv x_1^2 \equiv b \pmod{p},$$

thus

$$x_0^2 - x_1^2 = (x_0 - x_1)(x_0 + x_1) \equiv 0 \pmod{p}.$$

Hence, $p \mid (x_0 - x_1)$ or $p \mid (x_0 + x_1)$, which means $x_0 \equiv x_1 \pmod{p}$ or $x_0 \equiv -x_1 \pmod{p}$. Therefore, if there is a solution, there are exactly two incongruent solutions. □

Quadratic Residues (二次剩余) and Quadratic Non-Residue (二次非剩余) .

Definition

Let $m, n \in \mathbb{Z}$ with $\gcd(m, n) = 1$. Then m is called a *quadratic residue modulo n* (模 n 的二次剩余) (short for **QR**) if $m \equiv x^2 \pmod{n}$ for some $x \in \mathbb{Z}$, and m is called a *quadratic nonresidue modulo n* (模 n 的二次非剩余) (short for **QNR**) otherwise.

Example

From these tables we see that, the **QR** modulo 7 are $\{1, 2, 4\}$; the **QR** modulo 13 are $\{1, 3, 4, 9, 10, 12\}$; the **QNR** modulo 7 are $\{3, 5, 6\}$; the **QNR** modulo 13 are $\{2, 5, 6, 7, 8, 11\}$.

Theorem

*Let p be an odd prime, then there are exactly $(p - 1)/2$ **QRs** modulo p and exactly $(p - 1)/2$ **QNRs** modulo p .*

Proof.

We map the integers from $\mathbb{Z}_p^*$ to **QRs** modulo p by squaring. Since Proposition 1 tells us that, each **QR** has exactly two preimages under the squaring map. We have $p - 1$ squares to consider, then there are exactly $(p - 1)/2$ **QRs** modulo p . The remaining $(p - 1)/2$ integers are **QNRs** modulo p . □

Quadratic Residues Group.

Proposition

*Let $\mathbb{QR}_p$ denote the set of every **QR** modulo p , $\mathbb{QR}_p$ is a group under multiplication.*

Proof.

It is easy to prove by checking all the group axioms.



Quadratic Residues Group.

Remark.

The map $\phi : \mathbb{Z}_p^* \rightarrow \mathbb{QR}_p$ is a group homomorphism. From earlier observation we know $\ker \phi = \{1, p-1\}$. By the First Isomorphism Theorem from Chapter 9, this gives another proof of the previous theorem.

Quadratic Residues vs Quadratic Non-Residues.

Proposition

Let p be an odd prime. Then

*(1.) The product of two **QR**s modulo p is a **QR**, denoted as*

$$\mathbf{QR} \times \mathbf{QR} = \mathbf{QR},$$

*(2.) The product of a **QR** modulo p and a **QNR** modulo p is a **QNR** modulo p , denoted as*

$$\mathbf{QR} \times \mathbf{QNR} = \mathbf{QNR},$$

*(3.) The product of two **QNR**s modulo p is a **QR**, denoted as*

$$\mathbf{QNR} \times \mathbf{QNR} = \mathbf{QR}.$$

Quadratic Residues vs Quadratic Non-Residues.

Proof.

$$\mathbf{QR} \times \mathbf{QNR} = \mathbf{QNR},$$

Let b be a **QNR**, and a be a **QR**, then there exists $a_1 \in \mathbb{Z}$ such that $a \equiv a_1^2 \pmod{p}$. Assume that ab is a **QR**, then there exists $c \in \mathbb{Z}$ such that

$$c^2 \equiv ab \equiv a_1^2 b \pmod{p}.$$

We know that $\gcd(a_1, p) = 1$, then there exists $a_1^{-1} \in \mathbb{Z}$ such that $a_1 a_1^{-1} \equiv 1 \pmod{p}$, therefore we have

$$c^2 (a_1^{-1})^2 \equiv b \pmod{p}.$$

This means that b is a **QR** contradicting the assumption that b is a **QNR**. □

Quadratic Residues vs Quadratic Non-Residues.

Proof.

$$\mathbf{QNR} \times \mathbf{QNR} = \mathbf{QR}.$$

Let a be a **QNR** and $p \nmid a$, then we know

$$a\mathbb{Z}_p^* = \mathbb{Z}_p^*.$$

We also know that there are $(p-1)/2$ **QRs** and $(p-1)/2$ **QNRs** in $\mathbb{Z}_p^*$. As we have proved, when we multiply a by a **QR** we get a **QNR**, hence the $(p-1)/2$ products $a \times \mathbf{QR}$ give all $(p-1)/2$ **QNRs** in $\mathbb{Z}_p^*$. Then when we multiply a by a **QNR**, the only possibility is that it gives one of the **QRs** in $\mathbb{Z}_p^*$.



以上证明的代数视角.

使用群论理解二次剩余.

给定素数 p , 定义 $\mathbb{Z}_p^*$, 然后, $\mathbb{QR}_p$ 是 $\mathbb{Z}_p^*$ 的子群. 我们可以做商群 $\mathbb{Z}_p^*/\mathbb{QR}_p$, 这个商群只有两个元素, $\mathbb{QR}_p$ 是商群的单位元, $\mathbb{QNR}$ 则是该商群中的另一个元素. 分别把这个两个元素记为 $\{e, a\}$ 于是, 我们问:

- 为什么 $\mathbb{QR} \times \mathbb{QR} = \mathbb{QR}$? 答: $e * e = e$.
- 为什么 $\mathbb{QR} \times \mathbb{QNR} = \mathbb{QNR}$? 答: 因为 $e * a = a$.
- 为什么 $\mathbb{QNR} \times \mathbb{QNR} = \mathbb{QR}$? 答: $a * a$ 只能等于 e .

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- 为什么 $\mathbb{QNR} \times \mathbb{QNR} = \mathbb{QR}$? 答: $a * a$ 只能等于 e .

Remark.

一种更常见、更直观的记法是把群 $\{e, a\}$ 记为乘法群 $\{1, -1\}$. 即把 $\mathbb{QR}$ 记为 1, 把 $\mathbb{QNR}$ 记为 -1 . 请读者自行验证.

二次剩余的初始目标.

注记.

研究二次剩余的一个初始目标就是以求解线性同余方程的方法为基础, 试图解二次同余方程。即给定奇素数 p 和 $a, b, c \in \mathbb{Z}$, 其中要求 $\gcd(a, p) = 1$, 求解

$$ax^2 + bx + c \equiv 0 \pmod{p}$$

Legendre symbol. (勒让德符号)

Definition

Let p be an odd prime and let $n \in \mathbb{Z}$. The **Legendre symbol** (n/p) is defined as

$$\left(\frac{n}{p}\right) = \begin{cases} 1 & \text{if } n \text{ is a quadratic residue mod } p \\ -1 & \text{if } n \text{ is a quadratic nonresidue mod } p \\ 0 & \text{if } p \mid n. \end{cases}$$

Legendre symbol 的属性.

Proposition

设 p 是奇素数, $a, b \in \mathbb{Z}$ 且不被 p 整除。则有:

- ① 如果 $a \equiv b \pmod{p}$, 则 $\left(\frac{a}{p}\right) = \left(\frac{b}{p}\right)$;
- ② $\left(\frac{a}{p}\right) \left(\frac{b}{p}\right) = \left(\frac{ab}{p}\right)$;
- ③ $\left(\frac{a^2}{p}\right) = 1$ 。

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- ① 如果 $a \equiv b \pmod{p}$, 则 $\left(\frac{a}{p}\right) = \left(\frac{b}{p}\right)$;
- ② $\left(\frac{a}{p}\right) \left(\frac{b}{p}\right) = \left(\frac{ab}{p}\right)$;
- ③ $\left(\frac{a^2}{p}\right) = 1$ 。

Proof.

This follows directly from Proposition 6.



Example

$$\left(\frac{75}{97}\right) = \left(\frac{3 \cdot 5 \cdot 5}{97}\right) = \left(\frac{3}{97}\right) = 1$$

Since $10^2 \equiv 3 \pmod{97}$, so 3 is a **QR**.

Legendre symbol.

Example

$$\left(\frac{75}{97}\right) = \left(\frac{3 \cdot 5 \cdot 5}{97}\right) = \left(\frac{3}{97}\right) = 1$$

Since $10^2 \equiv 3 \pmod{97}$, so 3 is a **QR**.

Example

$$\left(\frac{269}{97}\right) = \left(\frac{2 \cdot 97 + 75}{97}\right) = \left(\frac{75}{97}\right) = 1.$$

Euler's Criterion. (欧拉准则)

Theorem

(Euler's Criterion.) Let p be an odd prime and let $a \in \mathbb{Z}$ with $\gcd(a, p) = 1$. Then

$$\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}.$$

Euler's Criterion.

Proof.

We have two cases to consider. For the first case, we assume $\left(\frac{a}{p}\right) = 1$, which means that there exists $x \in \mathbb{Z}$ such that $x^2 \equiv a \pmod{p}$. By Fermat's little theorem, we have that

$$x^{(p-1)} \equiv 1 \pmod{p},$$

and

$$x^{(p-1)} \equiv (x^2)^{(p-1)/2} \equiv a^{(p-1)/2} \pmod{p}.$$

Hence

$$a^{(p-1)/2} \equiv 1 \equiv \left(\frac{a}{p}\right) \pmod{p}.$$



Euler's Criterion.

Proof.

For the second case, we assume $\left(\frac{a}{p}\right) = -1$, which means that the congruence $x^2 \equiv a \pmod{p}$ has no solutions. Using Fermat's little theorem again, we know that

$$0 \equiv a^{(p-1)} - 1 \equiv (a^{(p-1)/2} - 1)(a^{(p-1)/2} + 1) \pmod{p}.$$

Since $\left(\frac{a}{p}\right) = -1$, $a^{(p-1)/2} - 1 \not\equiv 0 \pmod{p}$, hence, it must be

$$a^{(p-1)/2} + 1 \equiv 0 \pmod{p}.$$

Therefore

$$a^{(p-1)/2} \equiv -1 \equiv \left(\frac{a}{p}\right) \pmod{p}$$



Example

$$\left(\frac{3}{97}\right) = 3^{(97-1)/2} \bmod 97 = 3^{48} \bmod 97 = 1.$$

Thus 3 is a **QR** mod 97.

Euler's Criterion: Finding the patterns.

Euler's Criterion.

Let $p = 7$, and for every $1 \leq a < p$, compute $a^i \bmod p$, where $1 \leq i < p$. We have:

a	a^2	a^3	a^4	a^5	a^6
1	1	1	1	1	1
2	4	1	2	4	1
3	2	6	4	5	1
4	2	1	4	2	1
5	4	6	2	3	1
6	1	6	1	6	1

总结第三列的规律，并解释，为什么？

Euler's Criterion.

Theorem

Let p be an odd prime then

$$\left(\frac{-1}{p}\right) = \begin{cases} 1 & \text{if } p \equiv 1 \pmod{4}; \\ -1 & \text{if } p \equiv -1 \pmod{4}. \end{cases}$$

Proof.

It is easy by using Euler's criterion.



Theorem about $\left(\frac{2}{p}\right)$.

Theorem

Let p be an odd prime. Then

$$\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8}.$$

Proof.

It is NOT obvious by using Euler's criterion.



Theorem about primes of the form $4k + 1$.

Proposition

模 4 余 1 的素数有无穷多。

Proof.

假设模 4 余 1 的素数有限，枚举之 $S = \{1, p_1, p_2, \dots, p_n\}$ 。令

$$N = (2p_1p_2 \dots p_n)^2 + 1 \quad (1)$$

显然， N 是一个奇数（注意上式构造中嵌入的那个 2），所以必然存在一个奇素数 p 使得 $p \mid N$ 。也就是说：

$$(2p_1p_2 \dots p_n)^2 \equiv -1 \pmod{p}$$

即 $\left(\frac{-1}{p}\right) = 1$ 。根据勒让德符号的属性，可知 p 是形如 $4k + 1$ 的素数，所以 $p \in S$ 。这说明 $p \mid (N - (2p_1p_2 \dots p_n)^2)$ ，即 $p \mid 1$ ，矛盾！□

Quadratic Reciprocity(二次互反律).

注记.

著名的定理二次互反律是一个优美且高效实用的定理。据说，德国数学天才高斯 18 岁时第一次给出了二次互反律的严格证明，他称二次互反律是算术理论中的宝石，是黄金定律。希望大家能理解它并会使用它。

Quadratic Reciprocity.

Theorem

(Quadratic Reciprocity, Version 1.) Let p, q be distinct odd primes.

$$\left(\frac{-1}{p}\right) = \begin{cases} 1 & \text{if } p \equiv 1 \pmod{4}; \\ -1 & \text{if } p \equiv -1 \pmod{4}. \end{cases}$$

$$\left(\frac{2}{p}\right) = \begin{cases} 1 & \text{if } p \equiv 1 \text{ or } 7 \pmod{8}; \\ -1 & \text{if } p \equiv 3 \text{ or } 5 \pmod{8}. \end{cases}$$

$$\left(\frac{q}{p}\right) = \begin{cases} \left(\frac{p}{q}\right) & \text{if } p \equiv 1 \pmod{4} \text{ or } q \equiv 1 \pmod{4}; \\ -\left(\frac{p}{q}\right) & \text{if } p \equiv q \equiv 3 \pmod{4}. \end{cases}$$

Quadratic Reciprocity.

Example

$$\left(\frac{14}{137}\right) = \left(\frac{2}{137}\right) \left(\frac{7}{137}\right) = \left(\frac{7}{137}\right) = \left(\frac{137}{7}\right) = \left(\frac{4}{7}\right) = 1.$$

Thus 14 is a **QR**.

Quadratic Reciprocity.

Theorem

(Quadratic Reciprocity, Version 2.) Let p, q be distinct odd primes. Then

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{((p-1)/2)((q-1)/2)}.$$

习题.

- 请计算以下勒让德符号： $\left(\frac{7}{11}\right)$, $\left(\frac{72}{131}\right)$, $\left(\frac{20}{23}\right)$, $\left(\frac{17}{31}\right)$ 。
- 设 $n = 7 \cdot 11$, 请手动计算求出同余方程 $x^2 \equiv a \pmod{n}$ 在 $a = 5$ 和 $a = 4$ 时的所有解。
- 设 p 是奇素数, 请证明 $\mathbb{Z}_p^*$ 的所有生成元都是模 p 的二次非剩余。
- 使用抽象代数的语言重新证明欧拉准则。